



Pneumonia in children >28 days old: Inpatient treatment

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INTRODUCTION

Community-acquired pneumonia (CAP) is defined as an acute infection of the pulmonary parenchyma in a patient who has acquired the infection in the community, as distinguished from hospital-acquired (nosocomial) pneumonia. CAP is a common and potentially serious illness with considerable morbidity.

The inpatient treatment of CAP and hospital-acquired pneumonia in children >28 days old will be reviewed here.

Our recommendations are generally consistent with practice guidelines provided by [the Pediatric Infectious Diseases Society/Infectious Diseases Society of America](#) and the British Thoracic Society [1,2]. (See '[Society guideline links](#)' below.)

Other topics related to CAP in children and management of pneumonia in neonates are discussed separately:

- (See "[Community-acquired pneumonia in children: Outpatient treatment](#)".)
- (See "[Pneumonia in children: Epidemiology, pathogenesis, and etiology](#)".)
- (See "[Community-acquired pneumonia in children: Clinical features and diagnosis](#)".)
- (See "[Neonatal pneumonia](#)".)

TERMINOLOGY

This topic discusses treatment for three categories of community-acquired pneumonia (CAP). These categories are defined as follows:

- **Nonsevere uncomplicated pneumonia** – In this context, nonsevere pneumonia refers to pneumonia that requires hospitalization for one or more indications (eg, age, level of respiratory distress, inability of the caregiver to care for the child at home) (table 1). This category of treatment is only intended for children without pneumonia complications. (See 'Nonsevere uncomplicated pneumonia' below.)
- **Severe uncomplicated pneumonia** – Children with severe pneumonia are in moderate to severe distress (eg, nasal flaring, grunting, altered mental status, peripheral oxygen saturation [SpO₂] <90 percent in room air, capillary refill ≥2 seconds) (table 2). This category of treatment is only intended for children without pneumonia complications. (See 'Severe uncomplicated pneumonia' below.)
- **Complicated pneumonia** – Complicated pneumonia refers to pneumonia associated with parapneumonic effusion/empyema, necrotizing pneumonia, or lung abscess. Complicated pneumonia can be associated with either nonsevere or severe clinical manifestations. Complicated pneumonia requires broader antimicrobial therapy than is used for uncomplicated pneumonia. It also may require invasive interventions to drain effusions or remove necrotic tissue. (See 'Complicated pneumonia' below and "Management and prognosis of parapneumonic effusion and empyema in children".)

INDICATIONS FOR HOSPITALIZATION

- **General indications for hospital admission** –
The decision to hospitalize a child with pneumonia is individualized based on age, underlying medical problems, and severity of illness. Indications for hospitalization include (table 1) [1-3]:
 - Young age – Hospitalization is often warranted for infants <3 to 6 months of age, unless a viral etiology or *Chlamydia trachomatis* is suspected and the infant is minimally symptomatic and not hypoxemic.
 - Hypoxemia (peripheral oxygen saturation [SpO₂] <90 percent in room air).
 - Dehydration or inability to maintain hydration orally; inability to feed in an infant.

Indications for inpatient management of community-acquired pneumonia in children

Indication	Comment
Most children <3 to 6 months of age	Exception: a viral etiology or <i>Chlamydia trachomatis</i> is suspected and the child is relatively asymptomatic without hypoxia
Inability to care for at home	Any child whose caregiver(s) cannot provide appropriate care or follow the management instructions
Hypoxia	SpO ₂ <90% on room air
Dehydration	Inability to maintain oral hydration or inability to feed an infant
Moderate to severe respiratory distress	Examples: RR >70 breaths per minute for infants <12 months of age, RR >50 breaths per minute for children ≥12 months of age, retractions, nasal flaring, difficult breathing, apnea, grunting
Toxic appearance	Toxic appearance is more common in bacterial pneumonia and may suggest more severe illness
Underlying conditions that predispose to a more severe course	Examples: cardiopulmonary disease, genetic syndromes, neurocognitive disorders
Underlying conditions that may be exacerbated by pneumonia	Examples: metabolic disorder
Underlying conditions that may adversely affect response to treatment	Examples: immunocompromised host
Complications of pneumonia present	Examples: effusion/empyema, necrotizing process, lung abscess
Suspicion or confirmation of a pathogen that has increased virulence	Examples: <i>Staphylococcus aureus</i> , group A streptococcus
Failure of outpatient therapy	Continued lack of response within 48 to 72 hours of antibiotic change

Indications for inpatient management of community-acquired pneumonia in children

Table 1 - larger image below

- Moderate to severe respiratory distress – This includes tachypnea (respiratory rate >70 breaths per minute [bpm] for infants <12 months of age and >50 bpm for older children), retractions, nasal flaring, difficulty breathing, apnea, and/or grunting.
- Toxic appearance – Toxic appearance is more common in bacterial pneumonia and may suggest a more severe course [4].
- Underlying conditions that may predispose to a more serious course of pneumonia (eg, heart disease, lung disease, immune deficiency, genetic syndromes, neuromuscular disorders) or may be worsened by pneumonia (eg, metabolic disorder).
- Complications (eg, parapneumonic effusion/empyema, necrotizing pneumonia, lung abscess). (See '[Complicated pneumonia](#)' below and '[Management and prognosis of parapneumonic effusion and empyema in children](#)'.)
- Suspected or confirmed pathogen associated with high likelihood of causing severe and/or rapidly progressive infection (eg, *Staphylococcus aureus* or group A *Streptococcus* [GAS]).
- Failure of outpatient therapy (worsening or lack of response within 48 to 72 hours of antibiotic change).

Most children who require hospitalization for pneumonia can be managed in the general inpatient pediatric unit. Indications for intensive care unit admission are outlined below.

- **Pediatric intensive care unit (PICU) admission** – Admission to the PICU is generally warranted for children with any of the following [1,2]:
 - Severe respiratory distress (eg, marked retractions, dyspnea, nasal flaring, grunting)
 - Need for advanced respiratory support (mechanical ventilation, noninvasive ventilation [NIV])
 - Moderate to severe hypoxemia (eg, requiring a high fraction of inspired oxygen [FiO₂] to maintain SpO₂ >92 percent [eg, FiO₂ >0.5])
 - Recurrent episodes of apnea
 - Depressed or altered mental status
 - Hemodynamic instability or shock (eg, progressive tachycardia, lactic acidosis, and/or hypotension despite fluid resuscitation)
 - Underlying medical condition that places the child at high risk for severe/complicated illness (eg, sickle cell disease, primary immune deficiency)

INFECTION CONTROL MEASURES

Rapid diagnostic tests — If possible, rapid diagnostic tests should be performed at the time of admission to facilitate patient cohorting and other infection control decisions. We prefer rapid diagnostic tests to cultures because of their general availability, and their results are reported sooner. (See "[Community-acquired pneumonia in children: Clinical features and diagnosis](#)", section on 'Rapid diagnostic testing'.)

General measures

- **Standard precautions** – Standard precautions include hand hygiene and, if exposure to infectious material is anticipated (eg, purulent wound, respiratory secretions), the use of personal protective equipment. Hand hygiene is the single most important measure to prevent the spread of infection. Guidelines for hand hygiene in health care settings can be accessed through the [Centers for Disease Control and Prevention](#).
- **Additional infection control measures** – Additional infection control measures for suspected respiratory pathogens depend on the likely pathogen. However, we favor simplifying the approach by placing all patients with suspected respiratory tract infection on both contact and droplet precautions. If measles or varicella are suspected, airborne precautions are also indicated.

Once the results of rapid diagnostic tests or cultures are available, measures can be narrowed as follows [5,6]:

- Respiratory syncytial virus (RSV), parainfluenza, and human metapneumovirus – Contact precautions (ie, gown and gloves) and mask according to standard precautions
- Influenza, rhinovirus, group A *Streptococcus* (GAS; first 24 hours of treatment), and *Mycoplasma pneumoniae* – Droplet precautions (ie, mask within 3 feet)
- Adenovirus – Contact and droplet precautions
- Measles – Airborne precautions (ie, use of fit-tested N95 mask)
- Varicella – Airborne and contact precautions
- Methicillin-susceptible *S. aureus* (MSSA) – Standard precautions
- Methicillin-resistant *S. aureus* (MRSA) or other multidrug-resistant organisms – Contact and droplet precautions if exposed to respiratory secretions and use of dedicated patient equipment

Additional details about respiratory precautions are discussed separately. (See ["Infection prevention: Precautions for preventing transmission of infection"](#).)

PRETREATMENT EVALUATION

- **Laboratory evaluation** – The typical laboratory evaluation for children hospitalized with pneumonia includes a complete blood count (CBC) with differential, inflammatory markers, and serum electrolytes.
- **CBC** – When assessed in conjunction with other clinical, laboratory, and radiographic features, the white blood cell (WBC) count can sometimes be helpful in distinguishing between typical bacterial, atypical bacterial, and nonbacterial causes of pneumonia:
 - An elevated WBC count ($>15,000/\text{microL}$) is suggestive of a typical bacterial etiology [7]. However, children with *M. pneumoniae*, influenza, or adenovirus pneumonia may also have an elevated WBC count [8-10].
 - A normal or reduced WBC count suggests a nonbacterial etiology. However, a severely ill patient with bacterial pneumonia may present with leukopenia with a predominance of immature cells.
 - Peripheral eosinophilia is characteristic of afebrile pneumonia of infancy (typically caused by *C. trachomatis*). (See ["Chlamydia trachomatis infections in newborns and young infants"](#), section on 'Pneumonia'.)
- **Inflammatory markers** – Baseline inflammatory markers (eg, erythrocyte sedimentation rate [ESR], C-reactive protein [CRP], serum procalcitonin [PCT]) obtained at admission and followed by serial measurements during treatment may provide useful information about a patient's response to therapy [1,11-16]. They may also assist with clinical management decisions (eg, when to discontinue antimicrobial therapy). (See ["Duration of treatment"](#) below.)

Inflammatory marker levels do not reliably differentiate between typical bacterial, atypical bacterial, and viral pneumonia, but they may be helpful for assessing the likelihood of each. In a meta-analysis of observational studies in children, an elevated CRP and PCT performed better than an elevated WBC count in distinguishing bacterial from viral pneumonia; however, the sensitivity and specificity of approximately 70 and 65 percent, respectively, are insufficiently accurate to be used in isolation [17].

In a prospective, multicenter study of 532 children hospitalized with radiographically confirmed community-acquired pneumonia (CAP), median PCT values for typical bacterial etiologies were higher than for atypical bacterial and viral etiologies, but the ranges overlapped [18]. However, no child with a serum PCT <0.1 ng/mL had a typical bacterial

etiology, suggesting that this threshold may be helpful for identifying a low risk of typical bacterial CAP. CRP and PCT do not appear to be useful in predicting illness severity [19].

- **Serum electrolytes** – We suggest measuring serum electrolytes in all children requiring hospitalization for pneumonia since dehydration and hyponatremia are common in this setting.
- **Radiographic evaluation** – Children who are admitted to the hospital with pneumonia should have chest radiographs to confirm the diagnosis and assess for complications (eg, parapneumonic effusion). The radiographic evaluation is discussed in detail separately. (See "[Community-acquired pneumonia in children: Clinical features and diagnosis](#)", section on '[Chest imaging](#)'.)
- **Microbiologic testing** – Children who are admitted to the hospital for CAP should undergo microbiologic testing that includes rapid diagnostic testing (see '[Rapid diagnostic tests](#)' above), as well as blood cultures and a sputum (expectorated or induced) Gram stain and culture. The microbiologic evaluation is discussed in detail separately. (See "[Community-acquired pneumonia in children: Clinical features and diagnosis](#)", section on '[Microbiologic testing](#)'.)

Microbiologic testing is necessary in all hospitalized children with CAP since the clinical, laboratory, and radiologic features do not reliably distinguish between different etiologies. (See "[Pneumonia in children: Epidemiology, pathogenesis, and etiology](#)", section on '[Etiologic agents](#)'.)

SUPPORTIVE AND ADJUNCTIVE CARE

Antipyresis and analgesia — Fever and pleuritic chest pain should be treated with antipyretics and/or analgesics (eg, [acetaminophen](#), nonsteroidal anti-inflammatory drug [NSAID]). Fever can increase tachypnea and work of breathing in infants and children with pneumonia. Pleuritic chest pain can lead to shallow breathing and impaired ability to cough. Opioid analgesia does not play any role in this setting unless the child has a chest tube in place and experiences breakthrough pain while receiving parenteral [ketorolac](#).

Respiratory support — Children hospitalized with pneumonia should receive respiratory support as indicated by their clinical condition [1,2].

For children with hypoxemia, we suggest titrating supplemental oxygen therapy to maintain peripheral oxygen saturation (SpO₂) ≥92 percent. Other experts use different targets for supplemental oxygen. Gentle bulb suctioning of the nares may help reduce oxygen requirements in infants and children whose nares are blocked with secretions. Minimal

handling of the child may also reduce oxygen requirements. (See ["Continuous oxygen delivery systems for the acute care of infants, children, and adults"](#).)

Children with severe pneumonia may require advanced respiratory support with high-flow nasal cannula, noninvasive ventilation (NIV; including continuous positive airway pressure [CPAP] and bilevel positive airway support [BPAP]), or, rarely, intubation and mechanical ventilation. Details of these modalities are beyond the scope of this topic. They are discussed separately. (See ["High-flow nasal cannula oxygen therapy in children"](#) and ["Noninvasive ventilation for acute and impending respiratory failure in children"](#) and ["Initiating mechanical ventilation in children"](#).)

In children who are severely ill, it may be necessary to monitor carbon dioxide tension via blood gas analysis in addition to SpO₂ by oximetry. A capillary or venous blood sample may be used because evaluation of the partial pressure of carbon dioxide (PCO₂) does not require arterial blood. Hypercarbia is an important sign of impending respiratory failure, particularly in the young infant who is tiring but may have preserved oxygenation. Indications for carbon dioxide monitoring include signs that the child is tiring out (eg, less responsive, less active) and lack of improvement (eg, persistent severe retractions, persistent respiratory distress) after several hours of NIV.

Fluid management — Children who cannot maintain adequate fluid intake due to work of breathing, fatigue, or risk of aspiration require supplemental fluid therapy, which is usually provided with intravenous (IV) isotonic fluids [20]. Nasogastric (NG) tube hydration is usually avoided in a severely ill child because it may compromise breathing. When NG tube hydration is necessary, the smallest tube possible should be used [2]. (See ["Maintenance intravenous fluid therapy in children"](#).)

Children with pneumonia are at risk for syndrome of inappropriate antidiuretic hormone secretion (SIADH) [21,22]. Serum electrolytes, fluid balance, and urine specific gravity should be monitored if there is clinical suspicion of SIADH [22]. Confirmation of SIADH is discussed separately. (See ["Pathophysiology and etiology of the syndrome of inappropriate antidiuresis \(SIAD\)"](#), [section on 'Pulmonary disease'](#) and ["Maintenance intravenous fluid therapy in children"](#).)

Measures with unproven or no benefit

- **Antitussives** – Antitussives should be avoided, as none have been found to be effective in pneumonia [23]. Symptomatic treatment of cough is discussed separately. (See ["The common cold in children: Management and prevention"](#), [section on 'Cough'](#).)
- **Chest physiotherapy (CPT)** – Routine use of CPT does not appear to be beneficial for children with uncomplicated community-acquired pneumonia (CAP) [2]. A systematic review and meta-analysis identified six trials (559 patients, ages 29 days to 12 years) assessing the

effectiveness of various CPT interventions including conventional CPT with and without positive expiratory pressure, CPAP (nasal and bubble), and assisted autogenic drainage in children hospitalized with pneumonia [24]. Of the five trials reporting changes in SpO₂ after the CPT intervention, two trials reported improvements in SpO₂ whereas three trials did not detect any improvement. There was little to no difference in length of hospital stay between children who did or did not receive CPT (based on four trials).

- **Adjunctive glucocorticoid therapy** – We do not routinely provide glucocorticoid therapy to children hospitalized with pneumonia. Although some data suggest benefit, trials are limited, and more data are needed before this practice can be adopted as routine.

A 2017 systematic review and meta-analysis identified three small pediatric trials assessing the effect of systemic glucocorticoid therapy in children with *M. pneumoniae* pneumonia (two trials) or CAP (one trial) [25]. There were no deaths among the 225 children enrolled in these trials. The time to clinical cure was faster for children assigned to glucocorticoid therapy compared with those in the control group (mean difference 1.6 days, 95% CI 0.6-2.6 days). Children receiving glucocorticoid therapy were also less likely to have early clinical failure (defined as radiographic progression or clinical instability at day 5 through 8) compared with those in the control group (relative risk [RR] 0.41, 95% CI 0.24-0.70). However, the clinical failure rate in the control group was unusually high (66 percent) and not consistent with contemporary practice in resource-abundant settings. Thus, this finding may not be generalizable to other settings.

In a retrospective study of 2244 children treated for CAP in the outpatient setting, 13 percent received adjunctive glucocorticoid therapy, most of whom had underlying asthma [26]. Among children without underlying asthma, use of adjunctive glucocorticoids was associated with increased likelihood of having an unplanned acute care visit for respiratory complaints within 14 days of the CAP diagnosis. This finding suggests that glucocorticoid therapy may have a detrimental effect in children with nonsevere CAP in the absence of underlying asthma.

Additional large, well-designed pediatric trials are needed to clarify the role of adjunctive glucocorticoids in treating CAP.

EMPIRIC ANTIMICROBIAL THERAPY

Overview — Children who are hospitalized with pneumonia are initially treated empirically pending results of microbiologic testing. The choice of empiric therapy depends on the child's age, immunization status, severity of the infection, and

Age group and clinical presentation	Empiric antibiotic regimen	Comments
Infants < 6 months Moderate to severe pneumonia (a) > 3 days of illness (b) > 3 days of fever (c) > 3 days of cough (d) > 3 days of tachypnea (e) > 3 days of chest retractions	(a) Amoxicillin (b) Clindamycin (c) Ceftriaxone	(a) > 3 days of illness (b) > 3 days of fever (c) > 3 days of cough (d) > 3 days of tachypnea (e) > 3 days of chest retractions
Infants 6 months to 5 years Moderate to severe pneumonia (a) > 3 days of illness (b) > 3 days of fever (c) > 3 days of cough (d) > 3 days of tachypnea (e) > 3 days of chest retractions	(a) Amoxicillin (b) Clindamycin (c) Ceftriaxone (d) Azithromycin (e) Doxycycline	(a) > 3 days of illness (b) > 3 days of fever (c) > 3 days of cough (d) > 3 days of tachypnea (e) > 3 days of chest retractions
Children 6 years to 18 years Moderate to severe pneumonia (a) > 3 days of illness (b) > 3 days of fever (c) > 3 days of cough (d) > 3 days of tachypnea (e) > 3 days of chest retractions	(a) Amoxicillin (b) Clindamycin (c) Ceftriaxone (d) Azithromycin (e) Doxycycline	(a) > 3 days of illness (b) > 3 days of fever (c) > 3 days of cough (d) > 3 days of tachypnea (e) > 3 days of chest retractions

whether the pneumonia is uncomplicated or complicated (eg, parapneumonic effusion/empyema, necrotizing pneumonia, lung abscess) (☞ [table 3A](#) and ☞ [table 3B](#)). Factors that must be considered include the most likely pathogens, local antimicrobial susceptibility patterns, simplicity, tolerability, safety, and cost [27].

There are few randomized controlled trials to guide the choice of empiric antimicrobials in children with community-acquired pneumonia (CAP). The recommendations of most guidelines are based on in vitro susceptibilities of the most likely pathogens, rather than evidence of the superiority of one antimicrobial over another. Clinical response to empiric therapy and results of microbiologic studies, when available, help to determine whether additional evaluation or changes in therapy are necessary [1,2]. (See '[Pathogen-directed therapy](#)' below and '[Response to therapy](#)' below.)

Consultation with an infectious diseases specialist is advised for children with medication allergies, comorbid conditions, failure of outpatient therapy, or infection due to a multidrug-resistant organism. Consultation with a pediatric pulmonologist may also be helpful for children with recurrent pneumonia.

Nonsevere uncomplicated pneumonia

Etiologic clues to guide choice of empiric therapy

— In children with nonsevere uncomplicated CAP, the clinical presentation, laboratory findings, and radiographic findings can provide clues as to whether the pneumonia is more likely to be caused by a typical bacterial pathogen, an atypical bacterial pathogen, or a viral pathogen. These features or clues are summarized in the table (☞ [table 4](#)) and discussed in greater detail separately. (See "[Community-acquired pneumonia in children: Clinical features](#)"

Parenteral empiric antibiotics for inpatient treatment of pediatric community-acquired pneumonia[1,2]

Table 3a - larger image below

Doses for parenteral antibiotics for the empiric treatment of community-acquired pneumonia in hospitalized children

Table 3b - larger image below

and diagnosis", section on 'Bacterial versus viral etiology' and "Pneumonia in children: Epidemiology, pathogenesis, and etiology".)

Clinical and radiographic clues to the etiology of pneumonia in children*

The features often overlap, and they do not reliably distinguish between the different etiologies. Nevertheless, they can be helpful in guiding empiric therapy pending results of microbiologic testing.

Table 4 - larger image below

We suggest that clinicians use their best judgment as to whether the child's overall clinical picture (including the presentation, laboratory results, and radiographic findings) is most consistent with typical bacterial pneumonia, atypical bacterial pneumonia, a mixed picture, or a viral illness ([table 4](#)).

- If the clinical picture is most consistent with typical bacterial pneumonia (eg, lobar consolidation on chest radiograph with high fever and elevated inflammatory markers), empiric therapy should target common bacterial causes of CAP (eg, *Streptococcus pneumoniae*, group A *Streptococcus* [GAS], *Haemophilus influenzae*), as discussed below. (See 'Age-based regimens for nonsevere uncomplicated CAP' below.)
- If the clinical picture is most consistent with atypical bacterial pneumonia (eg, myalgias, conjunctivitis, sore throat, rash), empiric therapy consists of a macrolide (eg, [azithromycin](#)) to cover atypical bacterial pathogens (*M. pneumoniae*, *Chlamydia pneumoniae*).
- If the clinical picture is mixed, combination therapy is generally appropriate (ie, an agent that covers typical bacterial pathogens plus a macrolide to cover atypical pathogens).
- If the clinical picture is most consistent with a viral infection and there is no concern for bacterial coinfection, antibiotic therapy is not necessary. Antiviral therapy should be provided if a treatable viral etiology is identified (eg, influenza or coronavirus disease 2019 [COVID-19]). For other viral pathogens (eg, respiratory syncytial virus [RSV], rhinovirus, adenovirus), treatment is supportive. (See 'Viral pathogens' below and 'Confirmed viral etiology' below.)

While we suggest using this approach to help guide the initial choice of empiric therapy, clinicians should recognize that making these judgments is not an exact science. Therefore, it is important to regularly reevaluate the patient to assess their response to therapy and modify treatment if necessary. (See 'Response to therapy' below.)

Age-based regimens for nonsevere uncomplicated CAP

Infants aged 4 weeks to <6 months — For most infants ≥ 4 weeks and <6 months of age, we suggest treatment with [ceftriaxone](#) (or [cefotaxime](#), where available) ([table 3A](#) and

 [table 3B](#)). Infants at this age are not fully immunized against *S. pneumoniae* and *H. influenzae* type b (Hib). Third-generation cephalosporins provide empiric coverage for penicillin-resistant strains of *S. pneumoniae* (PRSP) and beta-lactamase-producing strains of invasive *H. influenzae* [4,28-30].

Afebrile well-appearing infants between the ages of two weeks and four months may have pneumonia caused by *C. trachomatis* (sometimes called afebrile pneumonia of infancy), which is characterized by insidious onset of nasal congestion and possibly conjunctivitis, followed by gradual development of staccato cough and tachypnea. *C. trachomatis* pneumonia is typically a mild illness, and most affected infants do not require hospitalization unless their illness is associated with dehydration or other complications. *C. trachomatis* pneumonia is treated with macrolide antibiotics (eg, [azithromycin](#)), as discussed separately. (See "[Chlamydia trachomatis infections in newborns and young infants](#)", section on 'Pneumonia'.)

Infants and children 6 months to <4 years — *S. pneumoniae* is the most common bacterial cause of pneumonia in this age group. Other bacterial etiologies include *Streptococcus pyogenes* (GAS), Hib (if not fully immunized), other typeable non-b and nontypeable *H. influenzae*, and *Moraxella catarrhalis* [2,4,28,31-35].

For infants ≥6 months and children <4 years, the choice of empiric antibiotic(s) depends on immunization status and community prevalence of PRSP. Local rates of PRSP are usually available from the antibiogram of the microbiology laboratory at the treating hospital.

Our suggested approach is as follows:

- For children with presumed typical bacterial CAP who are fully immunized ( [table 5](#)) and live in a community with rates of PRSP <25 percent, we suggest [ampicillin](#) or [penicillin G](#) [1,36,37]. The table provides dosing guidance ( [table 3B](#)).

For certain patients, oral [amoxicillin](#) may be a reasonable alternative, as discussed below. (See '[Parenteral therapy preferred for most patients](#)' below.)

- For children who are not fully immunized or live in a community with rates of PRSP ≥25 percent, we suggest treatment with [ceftriaxone](#) (or [cefotaxime](#), where available) rather than [ampicillin](#) or [penicillin G](#).
- For children with findings consistent with atypical pneumonia (eg, myalgias, conjunctivitis, sore throat, rash, sick contact within past month), we suggest adding a macrolide antibiotic (eg, [azithromycin](#)) to the empiric regimen. The incidence of *M. pneumoniae* in children hospitalized for pneumonia is rising [38].

- For patients with contraindications to penicillins (ie, history of immunoglobulin [Ig]E-mediated reaction or serious delayed reaction ([table 6](#))), [ceftriaxone](#) is an acceptable alternative ([table 3B](#)). For those with contraindications to beta-lactam antibiotics, [clindamycin](#) or [levofloxacin](#) are acceptable alternatives.

Based on the available data, there does not appear to be any advantage to using combined beta-lactam plus macrolide therapy in children without clinical features of atypical pneumonia [39]. A prospective population-based study of 1418 children hospitalized with radiographically confirmed CAP did not detect any differences in length of stay, intensive care admission, rehospitalization, or self-reported recovery in children who received combination therapy compared with those who received beta-lactam monotherapy [39].

Children ≥4 years — Atypical bacterial pathogens (eg, *M. pneumoniae*, *C. pneumoniae*) cause pneumonia more commonly in children ≥4 years old than in younger children. In addition, the typical bacterial CAP pathogens (*S. pneumoniae*, GAS, *H. influenzae*) are also common in this age group.

For children ≥4 years of age, the choice of empiric antibiotic(s) depends on immunization status, community rates of PRSP, and suspicion for atypical bacterial infection.

Our suggested approach is as follows:

- For children with presumed typical bacterial CAP who are fully immunized ([table 5](#)) and live in communities with a PRSP rate <25 percent, we suggest [ampicillin](#) or [penicillin G](#) ([table 3A-B](#)).

For certain patients, oral [amoxicillin](#) may be a reasonable alternative, as discussed below. (See '[Parenteral therapy preferred for most patients](#)' below.)

- For children who are not fully immunized or live in a community with rates of PRSP >25 percent, we suggest treatment with [ceftriaxone](#) (or [cefotaxime](#), where available) rather than [ampicillin](#) or [penicillin G](#).
- For children with findings consistent with atypical pneumonia (eg, myalgia, headache, photophobia, conjunctivitis, sore throat, rash, sick contact within past month), we suggest monotherapy with a macrolide ([table 3A-B](#)) rather than [ampicillin](#), [penicillin G](#), or a cephalosporin.
- For children with features of atypical pneumonia who also have strong evidence of a typical bacterial cause, such as white blood cell count (WBC) >15,000/microL, C-reactive protein (CRP) >35 to 60 mg/L (3.5 to 6 mg/dL), or chills, we suggest combination therapy with a macrolide plus a beta-lactam ([ampicillin](#), [penicillin G](#), or [ceftriaxone](#)) [4,40]. Use of a fluoroquinolone (eg, [levofloxacin](#), [moxifloxacin](#)) is a reasonable

alternative to cover both typical and atypical bacteria in older children and adolescents. In addition to its excellent gram-negative spectrum, levofloxacin is active against a number of other pathogens responsible for CAP, including beta-lactam-susceptible and nonsusceptible *S. pneumoniae*, *M. pneumoniae* (including macrolide-resistant *M. pneumoniae*), and *C. pneumoniae* [41]. However, levofloxacin-resistant *S. pneumoniae* have been identified [42].

- For children ≥ 4 years old and adolescents who have a contraindication to beta-lactam antibiotics (eg, history of IgE-mediated reaction or serious delayed reaction ([table 6](#))), [levofloxacin](#) is an acceptable alternative.

Parenteral therapy preferred for most patients — We suggest that children who require hospitalization for treatment of pneumonia be treated initially with parenteral antibiotics [1]. These children are typically not eating well and are receiving intravenous (IV) fluid hydration. In this setting, provide IV antibiotic therapy rather than trying to administer oral antibiotics to a sick child.

However, oral [amoxicillin](#) may be an acceptable alternative for infants and children who are fully immunized against Hib and *S. pneumoniae* and have nonsevere, uncomplicated pneumonia that is not thought to be due to *S. aureus* [43,44]. In a multicenter clinical trial in which patients with uncomplicated nonsevere CAP were randomly assigned to treatment with oral amoxicillin or IV [penicillin G](#), clinical outcomes were similar in both groups [45]. The British Thoracic Society guidelines also suggest that oral antibiotics are safe and effective for children hospitalized with pneumonia as long as they are able to tolerate oral fluids, are not vomiting, and do not have signs of septicemia or complicated pneumonia [2].

Suspicion for specific organisms

S. aureus — Children with pneumonia caused by *S. aureus* are typically toxic appearing with high fever and a severe, rapidly progressing illness. Staphylococcal pneumonia may be preceded by a viral respiratory tract infection (especially influenza) and may be associated with complications (eg, parapneumonic effusion/empyema, necrotizing pneumonia, lung abscess) [46,47]. For this reason, antistaphylococcal coverage is included in the empiric regimen for children with complicated pneumonia or severe pneumonia requiring intensive care management, as discussed below. (See '[Severe uncomplicated pneumonia](#)' below and '[Complicated pneumonia](#)' below.)

Additional guidance for selecting empiric antistaphylococcal coverage in children with suspected invasive *S. aureus* infections is summarized in the table ([table 7](#)) and discussed in detail separately. (See "[Staphylococcus aureus in children: Overview of treatment of invasive infections](#)", section on 'Choice of therapy'.)

Testing for nasal carriage of methicillin-susceptible *S. aureus* (MSSA)/methicillin-resistant *S. aureus* (MRSA) by culture or molecular tests may be helpful in reducing the use of anti-MSSA/MRSA therapy in the treatment of nonsevere uncomplicated pneumonia [48-50]. A negative result has a negative predictive value of approximately 95 percent for excluding MRSA pneumonia. Nevertheless, clinical judgment should be used to determine whether further testing is needed to completely rule out MRSA pneumonia.

Viral pathogens — Most children <5 years of age who are admitted to the hospital with pneumonia have viral pneumonia (eg, RSV, rhinovirus, adenovirus) [51]. This is particularly true in the absence of a lobar (or lobular) infiltrate and pleural effusion [4]. Viral pneumonia does not require empiric antibiotic therapy unless a mixed bacterial infection or secondary bacterial infection is suspected. Importantly, up to one-half of CAP in children may be of a mixed viral/bacterial etiology.

Common treatable viral causes of pneumonia include influenza and COVID-19. These are readily detectable with rapid diagnostic tests. (See '[Rapid diagnostic tests](#)' above and '[Confirmed viral etiology](#)' below.)

If a child is hospitalized during influenza season and results of rapid diagnostic testing for influenza are not readily available, empiric antiviral therapy (eg, [oseltamivir](#)) should be started pending laboratory confirmation. (See "[Seasonal influenza in children: Management](#)", section on '[Indications and preferred regimens](#)'.)

If influenza is complicated by secondary bacterial pneumonia, empiric antibiotic therapy should include coverage for *S. aureus*, including MRSA. Coinfection with influenza and *S. aureus* is particularly severe and can be rapidly fatal. (See "[Seasonal influenza in children: Clinical features and diagnosis](#)", section on '[S. pneumoniae, S. pyogenes, or S. aureus coinfection](#)'.)

Other considerations for antiviral therapy are discussed below. (See '[Confirmed viral etiology](#)' below.)

Aspiration pneumonia — For empiric treatment of aspiration pneumonia, we suggest [ampicillin-sulbactam](#) or [clindamycin](#) [52]. The former provides coverage for oral anaerobes as well as the usual bacterial causes of CAP (eg, *S. pneumoniae*, GAS, MSSA, *H. influenzae*). The latter's coverage is similar, except it lacks coverage against *H. influenzae*, and its GAS, *S. pneumoniae*, and MSSA coverage is not as broad.

[Clindamycin](#) is preferred if MRSA is suspected and the prevalence of clindamycin resistance in the community is <15 percent [53] or if the patient has a contraindication to beta-lactam antibiotics (eg, history of IgE-mediated reaction or severe delayed hypersensitivity reaction ([table 6](#))).

In neurologically compromised older adolescents prone to aspiration events, empiric treatment with a fluoroquinolone (eg, [moxifloxacin](#) 400 mg once daily) is a reasonable alternative. Moxifloxacin also has activity against anaerobic bacteria and both the usual typical and atypical bacterial causes of CAP.

Severe uncomplicated pneumonia — For children with severe pneumonia ([table 2](#)), we base the choice of empiric regimen on whether the child is critically ill (ie, requiring admission to the pediatric intensive care unit [PICU]). Indications for PICU care are discussed above. (See '[Indications for hospitalization](#)' above.)

- **Severe CAP not requiring PICU admission** – For children with severe CAP ([table 2](#)) who do not require admission to the PICU, we suggest combination empiric therapy with a third-generation cephalosporin (eg, [ceftriaxone](#) or [cefotaxime](#)) **plus** a macrolide (eg, [azithromycin](#)) ([table 3A-B](#)). Combination therapy improves coverage for resistant organisms and mixed bacterial/atypical bacterial infections.

If there is concern for *S. aureus*, options include (see '[S. aureus](#)' above):

- Adding either [vancomycin](#) or [clindamycin](#) (depending upon local susceptibility patterns) to a combination of third-generation cephalosporin and macrolide **or**
- Combination therapy with [ceftaroline](#) plus [azithromycin](#)

If the child is hospitalized during influenza season, antiviral therapy for influenza (eg, [oseltamivir](#)) may also be warranted. If results of rapid diagnostic testing for influenza are not readily available, empiric therapy should be started pending laboratory confirmation. (See "[Seasonal influenza in children: Management](#)", section on '[Antiviral therapy](#)'.)

- **Severe CAP requiring PICU admission** – Critically ill children requiring PICU care for severe pneumonia require broad-spectrum empiric therapy that provides coverage for potential beta-lactam resistance and MRSA.

We suggest the following regimen ([table 3A-B](#)) [54-56]:

- [Vancomycin](#) ([table 8](#)) **plus**
- [Nafcillin](#) or [oxacillin](#) **plus**

Clinical features of mild pneumonia (all of the following)	Clinical features of severe pneumonia (1 or more of the following)
Mild or absent respiratory distress: • No to mild tachypnea for age* • Mild or absent retractions • No grunting • No nasal flaring • No apnea • Mild shortness of breath	Severe respiratory distress: • RR >70 breaths/minute for infants; RR >50 breaths/minute for older children • Severe suprasternal, intercostal, or subcostal retractions (<12 months) • Severe difficulty breathing (>12 months) • Greeting • Nasal flaring • Apnea • Significant shortness of breath
Normal color	Cyanosis
Normal mental status	Altered mental status
Normoxemia (oxygen saturation ≥90% in room air)	Hypoxemia (sustained oxygen saturation <90% in room air)
Normal feeding (infants); no vomiting	Not feeding (infants) or signs of dehydration (older children)
Normal heart rate	Tachycardia
Capillary refill <2 seconds	Capillary refill ≥2 seconds

Table 2 - larger image below

- [Ceftriaxone](#) (or [cefotaxime](#), where available) **plus**
- [Azithromycin](#)

The rationale for including both [vancomycin](#) and [nafcillin/oxacillin](#) in the empiric regimen is that nafcillin/oxacillin is more effective than vancomycin for treating MSSA. (See "[Staphylococcus aureus in children: Overview of treatment of invasive infections](#)", section on 'MSSA infection'.)

If the child is hospitalized during influenza season, antiviral therapy for influenza (eg, [oseltamivir](#)) may also be warranted. If results of rapid diagnostic testing for influenza are not readily available, empiric therapy should be started pending laboratory confirmation. (See "[Seasonal influenza in children: Management](#)", section on 'Antiviral therapy'.)

Critically ill children receiving [vancomycin](#) therapy should have kidney function tests and vancomycin levels monitored. The approach to therapeutic monitoring for vancomycin in children is discussed separately. (See "[Staphylococcus aureus in children: Overview of treatment of invasive infections](#)", section on 'Therapeutic monitoring for vancomycin'.)

If there are contraindications to the use of either [vancomycin](#) or [nafcillin/oxacillin](#) in the above regimen, [linezolid](#) is a reasonable alternative to the combination ([table 3B](#)). Linezolid is an oxazolidinone antibiotic with activity against gram-positive cocci, including beta-lactam-resistant *S. pneumoniae* and MRSA; however, it is bacteriostatic rather than bactericidal against *S. aureus*.

An alternative non-vancomycin-containing regimen consists of [ceftaroline](#) and [azithromycin](#) ([table 3B](#)).

Complicated pneumonia

- **Parapneumonic effusion/empyema or necrotizing pneumonia** – Empiric antimicrobial therapy for these complications requires coverage for *S. pneumoniae*, GAS, and *S. aureus*, including MRSA.

We suggest the following regimen ([table 3B](#) and [table 8](#)):

- [Ceftriaxone](#) (or [cefotaxime](#), where available) **plus**
- [Clindamycin](#) or [vancomycin](#)

For patients with these complications in the setting of aspiration pneumonia, we use the same regimen as for aspiration pneumonia, as outlined above. (See '[Aspiration pneumonia](#)' above.)

Vancomycin is reserved for patients who are allergic to **clindamycin** or who are living in communities with a clindamycin-resistant *S. aureus* prevalence ≥ 15 percent (hospital thresholds vary). Additional considerations include community MRSA prevalence > 25 percent and severe illness. Kidney function and/or use of other nephrotoxic agents and turnaround time for susceptibility results also factor into the decision. Additional details, including vancomycin monitoring, are discussed above. (See '**Severe uncomplicated pneumonia**' above.)

Monotherapy with **ceftaroline** (see [table 3B](#)) is an alternative if there are contraindications to **penicillin G**, **ampicillin**, **clindamycin**, or **vancomycin**. In a multicenter randomized trial of 40 children with complicated community-acquired bacterial pneumonia (only one had documented MRSA), clinical cure rates at the end of treatment with ceftaroline monotherapy were similar to combination **ceftriaxone** and vancomycin (approximately 80 percent) [57]. (See "**Community-acquired pneumonia in children: Clinical features and diagnosis**", section on '**Complications**'.)

In addition to antibiotic therapy, children with moderate to large parapneumonic effusions or an empyema require invasive procedures to drain the pleural fluid. This is discussed separately. (See "**Management and prognosis of parapneumonic effusion and empyema in children**".)

- **Lung abscess** – Empiric antimicrobial therapy for a lung abscess in an otherwise healthy child requires coverage for the common typical bacterial etiologies of CAP (*S. pneumoniae*, GAS, *S. aureus*, *H. influenzae*) and oral anaerobes in case there was a preceding aspiration event.

We suggest one of the following regimens (see [table 3B](#)):

- **Ampicillin-sulbactam** or
- **Ceftriaxone** (or **cefotaxime**, where available) **plus clindamycin** (**vancomycin** is an alternative if the prevalence of clindamycin-resistant *S. aureus* is ≥ 15 percent in the community)

For patients with lung abscess in the setting of aspiration pneumonia, we use the same regimen as for aspiration pneumonia, as outlined above. (See '**Aspiration pneumonia**' above.)

Special situations

Hospital-acquired pneumonia — We suggest a broad-spectrum empiric antibiotic regimen for most patients with hospital-acquired pneumonia. These patients require coverage for gram-positive (eg, *S. aureus*), gram-negative (eg, Enterobacteriaceae, *Pseudomonas aeruginosa*), and anaerobic pathogens (if aspiration pneumonia is a possibility). We pair an

antipseudomonal beta-lactam antibiotic (for gram-negative pathogens) with another agent that covers gram-positive pathogens with or without anaerobic coverage.

- [Piperacillin-tazobactam](#) 300 mg/kg per day IV in divided four doses (maximum 12 g/day), **or**
- [Meropenem](#) 60 mg/kg per day IV in three divided doses (maximum 6 g/day), **or**
- [Cefepime](#) 150 mg/kg per day in three divided doses (maximum 4 g/day), if no suspicion of an aspiration event

[Vancomycin](#) or [linezolid](#) is added to the above regimen if MRSA is suspected based on known patient colonization or the epidemiology of the patient's unit ([table 3B](#) and [table 8](#)). (See '[S. aureus](#)' above.)

[Piperacillin-tazobactam](#) may not be preferred for gram-negative coverage if [vancomycin](#) is added because the combination of vancomycin and piperacillin-tazobactam has been associated with increased risk of acute kidney injury [58,59]. However, this risk must be weighed against the development of [meropenem](#) resistance, particularly in closed units.

If the patient is known to be colonized with a gram-negative organism that produces an extended-spectrum or AmpC beta-lactamase, we suggest therapy with [meropenem](#) alone. If the patient is neutropenic or in septic shock, we suggest the addition of [amikacin](#) 15 to 22.5 mg/kg per day divided in one or three divided doses.

Decisions about appropriate empiric antimicrobial therapy should be made in consultation with an infectious disease specialist.

Immunocompromised host — An aggressive approach to specific microbial diagnosis is indicated in immunocompromised hosts with clinically significant (eg, fever, signs of respiratory compromise) pneumonias. For patients with an endotracheal tube in place, specific microbial diagnosis may involve early flexible bronchoscopy for bronchoalveolar lavage with viral, fungal, and bacterial diagnostic studies. Although the protected specimen brush technique has been utilized in some settings, quantitative bacterial cultures are more commonly used to differentiate colonization from true lower respiratory tract infection. (See "[Flexible bronchoscopy in adults: Indications and contraindications](#)", section on 'Diagnostic indications' and "[Clinical presentation and diagnostic evaluation of ventilator-associated pneumonia](#)", section on 'Diagnostic evaluation' and "[Community-acquired pneumonia in children: Clinical features and diagnosis](#)", section on 'Sputum Gram stain and culture'.)

Decisions about appropriate empiric antimicrobial therapy should be made in consultation with an infectious disease specialist.

Empiric treatment for pneumonia in immunocompromised hosts requires broad-spectrum gram-positive and gram-negative coverage, similar to that required for hospital-acquired pneumonia. Immunocompromised patients are also at risk for *Pneumocystis jirovecii* (formerly *Pneumocystis carinii*). Common respiratory viruses (eg, influenza, RSV, adenovirus, severe acute respiratory syndrome coronavirus 2 [SARS-CoV-2]) may cause serious infections in immunocompromised children and may require antiviral treatment if a specific etiology is identified. (See '[Confirmed viral etiology](#)' below and "[Seasonal influenza in children: Management](#)", section on '[Antiviral therapy](#)' and "[Respiratory syncytial virus infection: Treatment in infants and children](#)", section on '[Immunocompromised patients](#)'.)

PATHOGEN-DIRECTED THERAPY

Once results of microbiologic tests are available, antimicrobial therapy can be directed toward the responsible pathogen or pathogens.

Confirmed viral etiology — The following viruses should be treated if identified:

- SARS-CoV-2 – Antiviral therapy is indicated if the child requires supplemental oxygen or other respiratory support. (See "[COVID-19: Management in children](#)", section on '[Decision to start pharmacotherapy](#)'.)
- Influenza – The diagnosis and treatment of influenza in children are discussed separately. (See "[Seasonal influenza in children: Management](#)", section on '[Antiviral therapy](#)' and "[Seasonal influenza in children: Clinical features and diagnosis](#)", section on '[Diagnosis](#)'.)
- Varicella zoster – [Acyclovir](#) is used in the treatment of varicella zoster pneumonia. (See "[Treatment of varicella \(chickenpox\) infection](#)".)

In immunocompromised children, antivirals and/or immunotherapy may be used when the following viruses are identified:

- Herpes simplex virus – [Acyclovir](#) can be used in the treatment of pneumonia due to herpes simplex virus (see "[Approach to the immunocompromised patient with fever and pulmonary infiltrates](#)".)
- Respiratory syncytial virus (RSV) – [Ribavirin](#) therapy with or without [nirsevimab](#) (see "[Respiratory syncytial virus infection: Treatment in infants and children](#)", section on '[Immunocompromised patients](#)'.)
- Parainfluenza – [Ribavirin](#) therapy (see "[Parainfluenza viruses in children](#)", section on '[Treatment](#)'.)

- Adenovirus – [Cidofovir](#) (see "[Diagnosis, treatment, and prevention of adenovirus infection](#)", section on 'Treatment')
- Cytomegalovirus (CMV) – [Ganciclovir](#) and [CMV immune globulin](#) (see "[Overview of cytomegalovirus \(CMV\) infections in children](#)", section on 'CMV infections in immunocompromised hosts')
- Other viral etiologies – Intravenous (IV) immunoglobulin (Ig)

Confirmed bacterial etiology — Directed antibiotic therapy for bacterial etiologies is as follows:

- Group A *Streptococcus* (GAS) – [Penicillin G](#) or [ampicillin](#) are the preferred parenteral agents for pneumonia caused by GAS ([table 3B](#)) [1].

Alternative parenteral agents include [cefazolin](#), [ceftriaxone](#), or [cefotaxime](#) (where available), [clindamycin](#) (if susceptible), and [vancomycin](#) (for children with contraindications to beta-lactam antibiotics [eg, history of IgE-mediated reaction or severe delayed hypersensitivity reaction ([table 6](#))] [1].

[Amoxicillin](#) or [penicillin V](#) are the preferred oral agents; the doses are as follows [1]:

- [Amoxicillin](#) 50 to 75 mg/kg per day orally divided in two doses (maximum 4 g/day)
- [Penicillin V](#) 50 to 75 mg/kg per day orally divided in three doses (maximum 2 g/day)

Alternative oral agents include [cephalexin](#) 75 to 100 mg/kg per day in three divided doses (maximum 4 g/day) or [clindamycin](#) (if susceptible) 40 mg/kg per day orally divided in three doses (maximum 1.8 g/day) [1].

- *S. pneumoniae* (see "[Pneumococcal pneumonia in children](#)", section on 'Specific therapy')
- *H. influenzae* (typeable or nontypeable) (see "[Epidemiology, clinical manifestations, diagnosis, and treatment of *Haemophilus influenzae*](#)", section on 'Directed treatment')
- *M. pneumoniae* (see "[Mycoplasma pneumoniae infection in children](#)", section on 'Management')
- *C. pneumoniae* (see "[Pneumonia caused by *Chlamydia pneumoniae* in children](#)")
- Methicillin-susceptible *S. aureus* (MSSA) – MSSA pneumonia may be treated with [oxacillin](#), [nafcillin](#), or [cefazolin](#) [1,4]
- Methicillin-resistant *S. aureus* (MRSA) (see "[Staphylococcus aureus in children: Overview of treatment of invasive infections](#)", section on 'Definitive antimicrobial therapy')

No pathogen identified — If diagnostic studies fail to identify an etiology of the pneumonia and the patient is clinically improving, transition to an oral antibiotic with the same spectrum of activity as the parenteral antibiotic that the patient is receiving.

ADDITIONAL CONSIDERATIONS

Duration of treatment

- **Parenteral antibiotic duration** – There are few data to guide decisions about the duration of parenteral therapy for community-acquired pneumonia (CAP) [2,60]. It is common to switch from parenteral antibiotics to oral therapy once the patient is clinically improving, afebrile for 24 to 48 hours, and not vomiting [61].
- **Total antibiotic duration** – The total duration of the antibiotic course is determined by the severity of the pneumonia, the causative agent, and the immune status of the host.

There are few randomized controlled trials available to guide decisions about the appropriate duration of antimicrobial therapy for radiographically confirmed childhood pneumonia that requires hospitalization [2].

- **Uncomplicated CAP** – For children hospitalized with uncomplicated CAP, we generally treat with a five- to seven-day course of combined parenteral and oral therapy.

This is supported by studies demonstrating the effectiveness of shorter courses of therapy for uncomplicated pneumonia [62-64]. In a single-center observational study in children ≥ 6 months of age who were hospitalized with uncomplicated CAP, rates of treatment failure were similar among those who received antibiotic therapy for 5 to 7 days and 8 to 14 days (3 versus 6 percent; odds ratio [OR] 0.48, 95% CI 0.18-1.20) [62]. Treatment failure was defined as a composite of unanticipated emergency department or outpatient visits, readmission, or death within 30 days after completion of antibiotics.

In a subsequent trial, patients with uncomplicated CAP who were randomly assigned to receive 5 days of antibiotics had a similar response to those assigned to receive 10 days of antibiotics [64]. Additionally, fewer patients who received the shorter course of antibiotics had antibiotic resistance genes detected on end-of-treatment throat swabs.

Randomized trials in adult patients with mild to moderate CAP also suggest that outcomes in patients treated for less than seven days are similar to those in patients treated longer. These trials are discussed separately. (See "[Treatment of community-acquired pneumonia in adults in the outpatient setting](#)", section on 'Duration of therapy'.)

- **Complicated pneumonia** – Treatment of complications (such as parapneumonic effusion/empyema, necrotizing pneumonia, and lung abscess) requires a prolonged

course of antibiotic therapy [65]. The duration is determined by the clinical response but is usually either a total of three to four weeks or another two weeks after the patient is afebrile and has improved clinically. Some infectious disease specialists use an erythrocyte sedimentation rate (ESR) value of <20 mm/hour as a laboratory indicator that the duration of therapy has been sufficient. (See '[Complicated pneumonia](#)' above.)

The duration of treatment for lung abscess is also determined by the radiographic resolution of the abscess or stabilization of a small lesion.

Response to therapy — The following clinical parameters should be monitored to assess response to treatment [1,2]:

- Temperature
- Respiratory rate
- Heart rate
- Peripheral oxygen saturation (SpO₂)
- Work of breathing (eg, retractions, nasal flaring, grunting)
- Mental status
- Ability to maintain oral intake and hydration
- Chest examination (extent of abnormal or absent breath sounds, extent of dullness to percussion)

The frequency of monitoring depends upon the severity of illness. In patients who are receiving oxygen therapy, SpO₂ should be evaluated regularly.

The respiratory status of children with uncomplicated CAP who are appropriately treated should improve within 48 to 72 hours [1]. However, fevers may persist for several days after initiation of appropriate therapy [52].

Treatment failure — In children with CAP who fail to improve as anticipated, the following possibilities must be considered [1,2,66,67]:

- Alternative or coincident diagnoses (eg, foreign body aspiration) (see "[Community-acquired pneumonia in children: Clinical features and diagnosis](#)", section on 'Differential diagnosis')
- Ineffective antibiotic coverage (lack of coverage for the actual etiology or resistant organism)

- Development of complications (see "[Community-acquired pneumonia in children: Clinical features and diagnosis](#)", section on 'Complications')
- Underlying immunodeficiency condition

The history should be reviewed with special attention to the possibility of foreign body aspiration and geographic or environmental exposures associated with pathogens not treated by the empiric regimen ([table 9](#)).

A repeat peripheral white blood cell (WBC) count and inflammatory markers should be sent because they may provide information about disease progression.

Repeat radiographs or additional imaging studies should be obtained to help assess the degree of parenchymal involvement and evaluate for complications or anatomic abnormalities [1].

If not already completed, a microbiologic evaluation should be done to help direct antibiotic therapy. (See "[Community-acquired pneumonia in children: Clinical features and diagnosis](#)", section on 'Microbiologic testing'.)

Invasive procedures may be necessary to obtain lower respiratory tract specimens for culture and other studies in children in whom an etiologic diagnosis is necessary and has not been established by other means. These procedures are typically reserved for seriously ill patients whose condition is worsening despite empiric therapy, individuals with significant comorbidities (eg, immune compromise), and those with complications [68-71] as follows:

- Bronchoscopy with bronchoalveolar lavage (BAL) – For samples obtained via bronchoscopy, quantitative culture techniques are suggested to differentiate true infection from upper airway contamination [72-74].
- Guided percutaneous needle aspiration or catheter drainage of a lung abscess – Microbiologic specimens may be obtained by ultrasonography or computed tomography-guided needle aspiration or catheter drainage. Ultrasonography is preferred because of the lack of radiation exposure.
- Lung biopsy, either by a thoracoscopic or thoracotomy approach – Samples obtained by lung biopsy often yield diagnostic information in children who had nondiagnostic BAL [75-77].

At least three weeks of intravenous (IV) antibiotic therapy should be delivered before lobectomy is considered for treatment failure [78]. Percutaneous drainage with catheter placement should be avoided, if possible, in children with necrotizing pneumonia. This procedure increases the risk of further complications, such as the development of bronchopleural fistulae [68,69,79,80].

Discharge criteria — Discharge criteria for children who have been admitted to the hospital with CAP have not been standardized but typically include [1,52]:

- Vital signs have improved
- Child is able to maintain adequate fluid and nutrition orally
- Improvement in respiratory status has improved
- SpO₂ is ≥90 percent in room air
- Overall clinical status has improved, including level of activity and appetite
- Afebrile or decreased fever for at least 12 to 24 hours
- Mental status is stable and at baseline
- Caregiver is able to administer home antibiotic regimen, and child is able to comply with therapy
- Child can be discharged to a safe and compliant home environment

Outpatient parenteral antimicrobial therapy (OPAT) is an option for selected patients who require prolonged parenteral treatment (usually for complicated CAP that for some reason cannot be treated with an oral antibiotic) and have stabilized clinically [52,81,82]. Eligibility for OPAT requires a suitable home environment and a pharmacologic agent with a reasonable dosing schedule [83]. Decisions regarding OPAT should involve the caregivers, an infectious diseases specialist (or clinician knowledgeable about the use of antimicrobial agents in OPAT), a hospital pharmacist, and the primary care provider. The services of a visiting nurse may be required for home visits, education of caregiver(s) and observation of caregiver administration, and/or obtaining blood samples for therapeutic monitoring.

Follow-up and prognosis

- **Follow-up**
 - **Primary care follow-up** – Children with pneumonia should be seen by their primary care provider soon after discharge to ensure that clinical improvement continues and antibiotic therapy is being taken as prescribed without ill effects [52]. Decisions regarding the timing of primary care follow-up should involve the child's primary care provider and include the clinical status of the child at the time of discharge.
 - **Resolution of symptoms** – Children who are appropriately treated for pneumonia should gradually improve with time. Cough may persist for as long as three to four months after viral pneumonia. Children who are recovering from typical or atypical bacterial pneumonia may continue to cough for several weeks and have moderate dyspnea on

exertion for two to three months [84]. Symptomatic treatment of cough is discussed separately. (See "[The common cold in children: Management and prevention](#)", section on 'Cough'.)

- **Radiographs** – Follow-up radiographs are not necessary in asymptomatic children with uncomplicated CAP, including round pneumonia [85]. However, in children with complicated pneumonia, particularly those who required invasive interventions, follow-up radiographs help to ensure resolution [2]. Follow-up radiographs may also be helpful in children with recurrent pneumonia, persistent symptoms, severe atelectasis, or unusually located infiltrates [2]. When follow-up radiographs are indicated, they should be obtained two to three weeks after hospital discharge.

Several studies have evaluated the utility of follow-up radiographs in cohorts of children with acute radiologically proven CAP [86-91]. Three of the studies included clinical as well as radiologic follow-up at three to seven weeks after initial diagnosis [86-89]. In each of these studies, follow-up radiographs were normal or improved in asymptomatic children. Residual radiographic findings, even when present, did not result in additional therapy.

Conditions that should be considered if symptoms persist in children with round pneumonia include congenital lung sequestration, pulmonary arteriovenous malformation, metastatic Wilms tumor, cavitary necrosis, pleural pseudocyst, and primary lung carcinoma [85,92-96].

- **Prognosis** – Most otherwise healthy children with pneumonia recover without sequelae, even if the pneumonia is complicated [52,79,80,97]. In a multicenter cohort study, approximately 3 percent of 82,566 children hospitalized with pneumonia were readmitted with pneumonia within 30 days of discharge; 8 percent were readmitted for any reason. Readmission was more common among children younger than one year of age and children with chronic medical conditions [98].

Although some data suggest that nearly one-half of children who are hospitalized for viral pneumonia have symptoms of asthma five years after hospitalization, it is not clear whether this is related to unrecognized asthma at the time of presentation with pneumonia or a causal relationship between community-acquired viral pneumonia and development of asthma [99,100].

The overall pneumonia mortality rate in resource-abundant countries is <1 per 1000 per year [51,101]. Pneumococcal pneumonia case fatality rates (not adjusted for comorbid conditions) for children in the United States were estimated to be 4 percent in children younger than two years and 2 percent in children 2 to 17 years before the introduction of pneumococcal conjugate vaccines (PCVs) [102].

The introduction of PCVs has resulted in a dramatic reduction (37 to 80 percent) in invasive disease and mortality rates in the countries in which they have been introduced [103]. However, pneumococcal pneumonia mortality rates have not been specifically examined. Data from the United States Pediatric Multicenter Pneumococcal Surveillance Study Group demonstrated all-cause pneumococcal mortality rates of 1 percent after the introduction of the seven-valent PCV (PCV7; during 2006 to 2009) and 0 percent after the introduction of the 13-valent PCV (PCV13; during 2011 to 2014) [104]. In a study from eastern Gambia, introduction of a nine-valent PCV was associated with a reduction in all-cause mortality from 30.1 to 25.2 per 1000 child-years [105]. (See "[Pneumococcal vaccination in children](#)", section on 'Efficacy and effectiveness'.)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "[Society guideline links: Pediatric pneumonia](#)".)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or email these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword[s] of interest.)

- Basics topic (see "[Patient education: Pneumonia in children \(The Basics\)](#)")

SUMMARY AND RECOMMENDATIONS

- **Indications for hospitalization** – For children with community-acquired pneumonia (CAP), decisions regarding the need for hospitalization are individualized based on age, underlying medical problems, and severity of illness. Indications for hospitalization are summarized in the table ([table 1](#)). (See '[Indications for hospitalization](#)' above.)

- **Pretreatment evaluation** – The pretreatment evaluation includes a complete blood count (CBC), inflammatory markers, serum electrolytes, chest radiographs, and a microbiologic evaluation. (See '[Pretreatment evaluation](#)' above.)
- **Infection control** – In addition to standard precautions, transmission-based infection control measures vary based on the microbial etiology of CAP. Rapid diagnostic tests may help with infection control decisions. (See '[Infection control measures](#)' above.)
- **Supportive care** – Supportive care for children hospitalized with pneumonia includes provision of antipyresis, analgesia, respiratory support, and fluid management. (See '[Supportive and adjunctive care](#)' above.)
- **Empiric antimicrobial therapy** – Children who are hospitalized with CAP are initially treated empirically pending results of microbiologic testing.

Our suggested approach is as follows ([table 3A-B](#)):

- **Infants ≥ 4 weeks to <6 months with nonsevere uncomplicated CAP** – For most infants ≥ 4 weeks to <6 months of age who are hospitalized with uncomplicated CAP, we suggest treatment with [ceftriaxone](#) or [cefotaxime](#) (where available) rather than other antibiotics (**Grade 2C**). Infants at this age are not fully immunized and are at risk for penicillin-resistant strains of *Streptococcus pneumoniae* (PRSP) and beta-lactamase-producing strains of *Haemophilus influenzae*, and ceftriaxone and cefotaxime provide excellent coverage. The table provides dosing guidance ([table 3B](#)).
- **Infants and children ≥ 6 months with nonsevere uncomplicated CAP** – Our suggested approach for infants and children ≥ 6 months of age is as follows (See '[Infants and children 6 months to <4 years](#)' above and '[Children \$\geq 4\$ years](#)' above.):
 - For children with presumed typical bacterial CAP who are fully immunized ([table 5](#)) and live in a community with rates of PRSP <25 percent, we suggest [ampicillin](#) or [penicillin G](#) rather than other antibiotics (**Grade 2C**).
 - For children who are not fully immunized ([table 5](#)) or live in a community with rates of PRSP ≥ 25 percent, we suggest [ceftriaxone](#) (or [cefotaxime](#), where available) rather than [ampicillin](#) or [penicillin G](#) (**Grade 2C**). (See '[Infants and children 6 months to <4 years](#)' above and '[Children \$\geq 4\$ years](#)' above.)
 - For children <4 years old who have findings consistent with atypical pneumonia (eg, myalgias, conjunctivitis, sore throat, rash), we suggest combination therapy with a macrolide (eg, [azithromycin](#)) plus a beta-lactam ([ampicillin](#), [penicillin G](#), or [ceftriaxone](#)) (**Grade 2C**). (See '[Infants and children 6 months to <4 years](#)' above.)

- For children ≥ 4 years old who have findings consistent with atypical pneumonia, we suggest monotherapy with a macrolide (eg, [azithromycin](#)) rather than combination therapy or beta-lactam monotherapy (**Grade 2C**). Children ≥ 4 years old are more commonly infected with atypical bacteria than are younger children and infants. (See '[Children \$\geq 4\$ years](#)' above.)
- For children with signs and symptoms consistent with both typical and atypical bacterial pneumonia, we suggest combination therapy with a macrolide (eg, [azithromycin](#)) plus a beta-lactam ([ampicillin](#), [penicillin G](#), or [ceftriaxone](#)) (**Grade 2C**).
- Alternative options for patients with penicillin or beta-lactam allergies are provided in the text above. (See '[Infants and children 6 months to <4 years](#)' above and '[Children \$\geq 4\$ years](#)' above.)
- **Aspiration pneumonia** – For children who are hospitalized for aspiration pneumonia, we suggest empiric treatment with [ampicillin-sulbactam](#) or [clindamycin](#) (**Grade 2C**). The former provides coverage for oral anaerobes as well as the usual bacterial causes of CAP (eg, *S. pneumoniae*, group A *Streptococcus* [GAS], methicillin-susceptible *Staphylococcus aureus* [MSSA], *H. influenzae*). The latter's coverage is similar except it lacks *H. influenzae* coverage, and its coverage against GAS, *S. pneumoniae*, and MSSA is not as broad. (See '[Aspiration pneumonia](#)' above.)
- **Severe and/or complicated pneumonia** – Our suggested approach to treating severe or complicated pneumonia in children is as follows:
 - **Severe uncomplicated CAP, not critically ill** – For children with severe uncomplicated CAP ([table 2](#)) who do not require admission to the pediatric intensive care unit (PICU), we suggest combination empiric therapy with a third-generation cephalosporin (eg, [ceftriaxone](#) or [cefotaxime](#)) plus a macrolide (eg, [azithromycin](#)) ([table 3A-B](#)). (See '[Severe uncomplicated pneumonia](#)' above.)
 - **Severe CAP, critically ill** – For critically ill children requiring PICU care for severe pneumonia, we suggest combination therapy with [ceftriaxone](#) (or [cefotaxime](#), where available) **plus** [vancomycin](#) **plus** either [nafcillin](#) or [oxacillin](#) **plus** [azithromycin](#) rather than other regimens (**Grade 2C**). Alternative options for patients with contraindications to vancomycin or beta-lactams are provided above. (See '[Severe uncomplicated pneumonia](#)' above.)
 - **Parapneumonic effusion/empyema or necrotizing pneumonia** – For patients with parapneumonic effusion/empyema or necrotizing pneumonia, we suggest combination therapy with [ceftriaxone](#) (or [cefotaxime](#), where available) **plus** either [clindamycin](#) or [vancomycin](#) rather than other regimens (**Grade 2C**). (See '[Complicated pneumonia](#)' above.)

- **Lung abscess** – For patients with lung abscess, we suggest either monotherapy with [ampicillin-sulbactam](#) or combination therapy with [ceftriaxone](#) (or [cefotaxime](#), where available) **plus** [clindamycin](#) ([vancomycin](#) is an alternative if the prevalence of clindamycin resistance in the community is ≥ 15 percent) rather than other regimens (**Grade 2C**). (See '[Complicated pneumonia](#)' above.)
- **Hospital-acquired pneumonia** – For patients with hospital-acquired pneumonia, we suggest empiric treatment with [piperacillin-tazobactam](#), [meropenem](#), or [cefepime](#) (**Grade 2C**). For children in whom methicillin-resistant *S. aureus* (MRSA) is suspected, we suggest the addition of [vancomycin](#) or [linezolid](#) to the regimen (**Grade 2C**). (See '[Hospital-acquired pneumonia](#)' above.)
- **Immunocompromised host** – Immunocompromised children who are hospitalized with pneumonia require broad-spectrum empiric coverage. Consultation with an infectious disease specialist is advised. (See '[Immunocompromised host](#)' above.)
- **Pathogen-directed therapy** – If microbiologic testing identifies a specific pathogen, antibiotic therapy is directed toward the specific pathogen. (See '[Pathogen-directed therapy](#)' above.)
- **Duration of treatment** – Parenteral therapy is typically transitioned to oral therapy when the patient has been afebrile for 24 to 48 hours and can tolerate oral intake. The total duration of antibiotic therapy is usually five to seven days for uncomplicated CAP. Up to four weeks of antimicrobial therapy may be necessary for complicated CAP. (See '[Duration of treatment](#)' above.)
- **Treatment failure** – The respiratory status of children receiving appropriate antimicrobial therapy for CAP should improve within 48 to 72 hours. Children who fail to improve as anticipated may not be receiving adequate antibiotic therapy, may have developed complications, or may have an alternative or coincident diagnosis. (See '[Treatment failure](#)' above.)
- **Follow-up and prognosis** – Children recovering from CAP may continue to cough for several weeks to four months, depending upon the etiology. Those recovering from typical or atypical bacterial pneumonia may also have moderate dyspnea on exertion for two to three months. After this recovery period, most otherwise healthy children do not have any long-term sequelae. (See '[Follow-up and prognosis](#)' above.)

Follow-up radiographs are not necessary in asymptomatic children with uncomplicated CAP, including round pneumonia. However, in children with complicated pneumonia, especially those who required intervention, follow-up radiographs help to ensure resolution. (See '[Follow-up and prognosis](#)' above.)

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Topic 6054 Version 73.0

GRAPHICS

Table 1: Indications for inpatient management of community-acquired pneumonia in children

Indication	Comment
Most children <3 to 6 months of age	Exception: a viral etiology or <i>Chlamydia trachomatis</i> is suspected and the child is relatively asymptomatic without hypoxia
Inability to care for at home	Any child whose caregiver(s) cannot provide appropriate care or follow the management instructions
Hypoxia	SpO ₂ <90% on room air
Dehydration	Inability to maintain oral hydration or inability to feed an infant
Moderate to severe respiratory distress	Examples: RR >70 breaths per minute for infants <12 months of age, RR >50 breaths per minute for children ≥12 months of age, retractions, nasal flaring, difficulty breathing, apnea, grunting
Toxic appearance	Toxic appearance is more common in bacterial pneumonia and may suggest more severe illness
Underlying conditions that predispose to a more severe course	Examples: cardiopulmonary disease, genetic syndromes, neurocognitive disorders
Underlying conditions that may be exacerbated by pneumonia	Example: metabolic disorder
Underlying conditions that may adversely affect response to treatment	Example: immunocompromised host
Complications of pneumonia present	Examples: effusion/empyema, necrotizing process, lung abscess
Suspicion or confirmation of a pathogen that has increased virulence	Examples: <i>Staphylococcus aureus</i> , group A <i>Streptococcus</i>
Failure of outpatient therapy	Continued lack of response within 48 to 72 hours of antibiotic change

We recommend that children with any of the above indications be hospitalized for management of CAP.

CAP: community-acquired pneumonia; RR: respiratory rate; SpO₂: peripheral capillary oxygen saturation.

Table 2: Severity of community-acquired pneumonia in infants and children

Clinical features of mild pneumonia (all of the following)	Clinical features of severe pneumonia (1 or more of the following)
Mild or absent respiratory distress: - No to mild tachypnea for age* - Mild or absent retractions - No grunting - No nasal flaring - No apnea - Mild shortness of breath	Severe respiratory distress: - RR >70 breaths/minute for infants; RR >50 breaths/minute for older children - Severe suprasternal, intercostal, or subcostal retractions (<12 months) - Severe difficulty breathing (≥12 months) - Grunting - Nasal flaring - Apnea - Significant shortness of breath
Normal color	Cyanosis
Normal mental status	Altered mental status
Normoxemia (oxygen saturation ≥90% in room air)	Hypoxemia (sustained oxygen saturation <90% in room air)
Normal feeding (infants); no vomiting	Not feeding (infants) or signs of dehydration (older children)
Normal heart rate	Tachycardia
Capillary refill <2 seconds	Capillary refill ≥2 seconds

RR: respiratory rate.

* World Health Organization definitions of tachypnea according to age are as follows: <2 months: >60 breaths/minute; 2 to 12 months: >50 breaths/minute; 1 to 5 years: >40 breaths/minute; >5 years: >20 breaths/minute.^[1]

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Adapted from:

- Bradley JS, Byington CL, Shah SS, et al. *The management of community-acquired pneumonia in infants and children older than 3 months of age: Clinical practice guidelines by the Pediatric Infectious Diseases Society and the Infectious Diseases Society of America*. *Clin Infect Dis* 2011; 53:e25.
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Graphic 72015 Version 8.0

Table 3A: Parenteral empiric antibiotics for inpatient treatment of pediatric community-acquired pneumonia

Age group and suspected pathogens	Suggested parenteral empiric agent(s)	Comments
1 to 6 months		
Bacterial (not <i>Chlamydia trachomatis</i> or <i>Staphylococcus aureus</i>)	1 of the following: - Ceftriaxone - Cefotaxime	- If MRSA is suspected, add 1 of the following: - Vancomycin or clindamycin - Ceftaroline* (alternative)
<i>C. trachomatis</i>	Azithromycin	
≥6 months		
Uncomplicated bacterial (not <i>Mycoplasma pneumoniae</i> , <i>Chlamydia pneumoniae</i> , or <i>S. aureus</i>)	1 of the following: - Ampicillin or penicillin G (preferred) - Cefotaxime - Ceftriaxone	- If MRSA is suspected, add 1 of the following: - Vancomycin or clindamycin[¶] - Ceftaroline* (alternative) - Cefotaxime and ceftriaxone are reserved for: - Children with incomplete Hib or <i>Streptococcus pneumoniae</i> immunizations, or - Communities with substantial prevalence of penicillin-resistant <i>S. pneumoniae</i> (eg, ≥25%)
<i>M. pneumoniae</i> or <i>C. pneumoniae</i>	1 of the following: ^Δ - Azithromycin - Erythromycin - Levofloxacin	
Clinical syndrome (any age)	Suggested empiric parenteral agent(s)	Comments
Severe pneumonia	Combination therapy with 1 of the following: - Ceftriaxone - Cefotaxime Plus 1 of the following: ^Δ - Azithromycin - Erythromycin	- Children with severe infection may benefit from broad-spectrum therapy that addresses both typical and atypical pathogens If <i>S. aureus</i> is a consideration, either:

	▪ Doxycycline	• Add vancomycin or clindamycin[¶], or • Provide therapy with ceftaroline* plus azithromycin
Severe pneumonia requiring ICU admission	Combination therapy with: ▪ Vancomycin Plus 1 of the following: ▪ Ceftriaxone ▪ Cefotaxime Plus:^Δ ▪ Azithromycin Plus: ▪ Antiviral treatment for influenza if the child is hospitalized during influenza season	▪ If <i>S. aureus</i> is likely: • Add nafcillin[◇], or • Substitute linezolid for vancomycin and nafcillin, or • Provide therapy with ceftaroline* plus azithromycin plus antiviral treatment for influenza if the child is hospitalized during influenza season
Complicated pneumonia (eg, effusion/empyema, necrotizing process, abscess [§])	Combination therapy with 1 of the following: ▪ Ceftriaxone ▪ Cefotaxime Plus: ▪ Clindamycin if <i>S. aureus</i> or anaerobic infection is a consideration	▪ Potential pathogens include <i>S. pneumoniae</i>, <i>S. aureus</i>, and <i>Streptococcus pyogenes</i> ▪ Vancomycin is an alternative to clindamycin for children with allergy to clindamycin or high prevalence of clindamycin resistance in the community[¶] ▪ Monotherapy with ceftaroline is an alternative if <i>S. aureus</i> is a consideration

This table is meant for use with UpToDate content on the treatment of CAP in children. Refer to related UpToDate content for details regarding complete Hib and *S. pneumoniae* immunization, criteria for severe pneumonia, and pneumonia requiring ICU admission. Consultation with a specialist in infectious diseases for children is suggested for children with severe hypersensitivity to beta-lactam antibiotics (eg, penicillins and cephalosporins).

CAP: community-acquired pneumonia; Hib: *Haemophilus influenzae* type b; ICU: intensive care unit; MRSA: methicillin-resistant *S. aureus*; MSSA: methicillin-susceptible *S. aureus*.

* Ceftaroline is a fifth-generation cephalosporin. It is available in the United States for the treatment of pediatric CAP due to *S. pneumoniae*, MSSA, and *H. influenzae* in children ≥2 months of age. Although ceftaroline has in vitro activity against MRSA, experience in children with documented MRSA CAP is limited.

¶ The threshold prevalence of clindamycin-resistant MRSA (constitutive plus inducible) for choosing vancomycin varies from center to center, usually ranging from 10 to 25%, in an effort to balance the benefit of definitive therapy for the patient with the risk of increasing vancomycin resistance in the

community. Additional considerations in the decision to choose vancomycin include the prevalence of MRSA in the community, the severity of illness, and the turn-around time for susceptibilities.

Δ If tolerated, oral therapy is recommended.

◇ Nafcillin is added if *S. aureus* is likely because MSSA is more rapidly killed by nafcillin than by vancomycin.

§ Ampicillin-sulbactam alone may be effective if lung abscess is thought to be secondary to aspiration.

Data from:

1. McIntosh K. Community-acquired pneumonia in children. *N Engl J Med* 2002; 346:429.
 2. Bradley JS, Byington CL, Shah SS, et al. The management of community-acquired pneumonia in infants and children older than 3 months of age: Clinical practice guidelines by the Pediatric Infectious Diseases Society and the Infectious Diseases Society of America. *Clin Infect Dis* 2011; 53:e25.
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Graphic 56260 Version 35.0

Table 3B: Doses for parenteral antibiotics for the empiric treatment of community-acquired pneumonia in hospitalized children

Agent	Regimen ^[1,2]	Comments
Ampicillin	150 to 200 mg/kg per day in 4 divided doses (maximum daily dose 12 g)	
Ampicillin-sulbactam	150 to 200 mg/kg per day of the ampicillin component IV in 4 divided doses (maximum daily dose 8 g) of the ampicillin component	
Azithromycin	10 mg/kg once per day on days 1 and 2 of therapy (maximum daily dose 500 mg) 5 mg/kg once per day on subsequent days of therapy (maximum daily dose 250 mg)	Transition to oral therapy at 5 mg/kg per day as soon as clinically appropriate
Cefazolin	100 to 150 mg/kg per day in 3 divided doses (maximum daily dose 12 g)	Limited data on doses >100 mg/kg per day
Cefotaxime	150 mg/kg per day in 3 or 4 divided doses (maximum daily dose 8 g)	The 4-dose regimen should be used for severe infection or substantial local penicillin resistance
Ceftaroline	- Age $\geq$ 2 months and <2 years: 8 mg/kg every 8 hours - Age $\geq$ 2 and <18 years: - Weight $\leq$ 33 kg: 12 mg/kg every 8 hours - Weight >33 kg: 400 mg every 8 hours or 600 mg every 12 hours - Age $\geq$ 18 years: 600 mg every 12 hours	Experience with ceftaroline in children with documented MRSA infection is limited
Ceftriaxone	50 to 100 mg/kg per day in 1 or 2 divided doses (maximum daily dose 4 g)	- The 100 mg/kg per day dose should be used only if local rates of penicillin resistance to <i>Streptococcus pneumoniae</i> are substantial (eg, $\geq$ 25%) - The 2-dose regimen should be used for severe infection or if local rates of penicillin resistance to <i>S. pneumoniae</i> are substantial (eg, $\geq$ 25%)

Clindamycin	30 to 40 mg/kg per day in 3 or 4 divided doses (maximum daily dose 2.7 g)	
Doxycycline	4.4 mg/kg per day in 2 divided doses (maximum daily dose 200 mg)	Transition to oral therapy as soon as clinically appropriate
Erythromycin	20 mg/kg per day in 4 divided doses (maximum daily dose 4 g)	Parenteral erythromycin is associated with phlebitis, prokinetic, and cardiotoxic effects (rare)
Levofloxacin	- Age 6 months and <5 years: 16 to 20 mg/kg per day in 2 divided doses - Age ≥5 to 16 years: 8 to 10 mg/kg once per day (maximum daily dose 750 mg)	- Fluoroquinolones may prolong QTc interval; avoid use in patients with:^[3] - Long QT syndrome - Hypokalemia or hypomagnesemia - Organic heart disease (eg, congestive heart failure; requiring a class Ia antiarrhythmic drug*, particularly quinidine) - Concurrent use of other medications that prolong the QTc interval
Linezolid	- Age <12 years: 10 mg/kg every 8 hours (maximum daily dose 600 mg/dose) - Age ≥12 years: 600 mg every 12 hours	
Nafcillin	150 to 200 mg/kg per day in 4 or 6 divided doses (maximum daily dose 12 g)	
Oxacillin	150 to 200 mg/kg per day in 4 or 6 divided doses (maximum daily dose 12 g)	
Penicillin G	For therapy: 200,000 to 250,000 units/kg per day in 4 or 6 divided doses (maximum daily dose 24 million units)	
Vancomycin	40 to 60 mg/kg per day in 3 or 4 divided doses (maximum daily dose 4 g)[¶]	

This table is meant for use with UpToDate content on the treatment of community-acquired pneumonia in children. Refer to related UpToDate content for details about choice of therapy.

Consultation with a specialist in pediatric infectious diseases is suggested for children with severe hypersensitivity to beta-lactam antibiotics (eg, penicillins and cephalosporins) and for immunocompromised children. The recommended doses are for children with normal kidney function.

IV: intravenous; MRSA: methicillin-resistant *Staphylococcus aureus*.

* Class Ia antiarrhythmic drugs include quinidine, ajmaline, disopyramide, and procainamide.

¶ Alternative dosing is suggested for clinicians/institutions who follow AUC-guided therapeutic monitoring for vancomycin for serious MRSA infections as suggested by [consensus guidelines](#)^[4]; this strategy requires input from a clinical pharmacist, who will provide recommendations for initial dosing. Refer to UpToDate content on invasive staphylococcal infections in children for details of trough-guided and AUC-guided vancomycin dosing.

References:

1. American Academy of Pediatrics. Tables of antibacterial drug dosages. In: *Red Book: 2021-2024 Report of the Committee on Infectious Diseases*, 32nd ed, Kimberlin DW, Barnett ED, Lynfield R, Sawyer MH (Eds), American Academy of Pediatrics, Itasca, IL 2021. p.876.
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Graphic 127763 Version 6.0

Table 4: Clinical and radiographic clues to the etiology of pneumonia in children

Etiology	Clinical features	Radiographic features
Bacterial (most commonly <i>Streptococcus pneumoniae</i>)	- Children of all ages - Abrupt onset - Ill appearance - Chills - Moderate to severe respiratory distress - Focal auscultatory findings - Localized chest pain - WBC count >15,000/mL (if obtained) - Elevated acute phase reactants (if obtained)	- Alveolar infiltrates - Segmental consolidation - Lobar consolidation "Round" pneumonia Complications: - Pleural effusion/empyema - Lung abscess - Necrotizing pneumonia - Pneumatocele
Atypical bacterial (most commonly <i>Mycoplasma pneumoniae</i>)	- Children of all ages (most common in children >5 years) - Abrupt onset with gradually worsening constitutional findings (malaise, myalgia, headache, rash, conjunctivitis, photophobia, sore throat) and nonproductive cough - Wheezing - Extrapulmonary manifestations or complications (eg, polymorphous mucocutaneous eruptions, hemolytic anemia, hepatitis, pancreatitis, myopericarditis, aseptic meningitis)	- Lobar or segmental consolidation (37%) - Perihilar or peribronchial infiltrates (27%) - Localized reticulonodular infiltrates (21%) - Patchy infiltrates (15%)
Viral	- Usually children <5 years - Gradual onset - Preceding upper airway symptoms - Nontoxic appearance - Diffuse, bilateral auscultatory findings - Wheezing - May have associated rash (eg, measles, varicella)	- Interstitial infiltrates - Associated bronchiolitis: - Patchy atelectasis - Peribronchial infiltrations with air bronchograms - Hyperinflation with flattening of the diaphragms

Afebrile pneumonia of infancy (most commonly <i>Chlamydia trachomatis</i>)	▪ Usually in infants 1 to 3 months of age ▪ Insidious onset ▪ Tachypnea, diffuse crackles ▪ Rhinorrhea ▪ Staccato cough pattern ▪ Peripheral eosinophilia (if CBC obtained)	▪ Hyperinflation with interstitial infiltrates
Fungal	▪ Appropriate geographic or environmental exposure	▪ Mediastinal or hilar adenopathy
<i>Mycobacterium tuberculosis</i>	▪ Children of any age ▪ Chronic cough ▪ Constitutional symptoms ▪ Exposure history	▪ Mediastinal or hilar adenopathy

CBC: complete blood count; WBC: white blood cell.

* The clinical features frequently overlap and cannot reliably distinguish between bacterial, atypical bacterial, and viral etiologies; up to one-half of community-acquired pneumonias in children may be mixed bacterial/viral infections. Chest radiography generally is not helpful in determining the potential causative agent of pneumonia. Nonetheless, these features may facilitate decisions regarding empiric therapy.

References:

1. Bartlett JG, Mundy LM. Community-acquired pneumonia. *N Engl J Med* 1995; 333:1618.
2. Boyer KM. Nonbacterial pneumonia. In: *Textbook of Pediatric Infectious Diseases*, 6th ed, Feigin RD, Cherry JD, Demmler-Harrison GJ, Kaplan SL (Eds), Saunders, Philadelphia 2009.
3. Broughton RA. Infections due to *Mycoplasma pneumoniae* in childhood. *Pediatr Infect Dis* 1986; 5:71.
4. McIntosh K. Community-acquired pneumonia in children. *N Engl J Med* 2002; 346:429.
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Graphic 52021 Version 18.0

Table 5: Suggested *Haemophilus influenzae* type b and *Streptococcus pneumoniae* immunization status when considering empiric antibiotics for full immunization for community-acquired pneumonia in children

Current age [*]	Criteria for full immunization [¶]
<i>Haemophilus influenzae</i> type b	
12 to 15 months	≥2 doses of Hib conjugate vaccine, with at least one dose at ≥12 months of age
15 months to 5 years	≥2 doses of Hib conjugate vaccine, with at least one dose at ≥12 months of age, or
	≥1 dose of Hib conjugate vaccine at ≥15 months of age
≥5 years, not high risk ^Δ	Hib immunization not necessary
<i>Streptococcus pneumoniae</i>	
12 to 24 months	≥3 doses of PCV at <16 months, with ≥1 dose at ≥12 months, or
	2 doses of PCV, both at ≥12 months (separated by 8 weeks or more)
24 months through 5 years	≥3 doses of PCV at <16 months, with ≥1 dose at ≥12 months, or
	2 doses of PCV, both at ≥12 months, or
	≥1 dose of PCV at ≥24 months
>5 years, not high risk [◇]	PCV immunization not necessary

Hib: *H. influenzae* type b; HIV: human immunodeficiency virus; PCV: pneumococcal conjugate vaccine.

* Children younger than 12 months are incompletely immunized against Hib and *S. pneumoniae*.

¶ Immunizations must be completed at least two weeks before pneumonia diagnosis.

Δ Children at high risk for invasive Hib disease include chemotherapy recipients and those living with anatomic or functional asplenia (including sickle cell disease), HIV infection, immunoglobulin deficiency, or early component complement deficiency. Please refer to the UpToDate topic on prevention of *H. influenzae* infection for a discussion of full Hib immunization in children at high risk for invasive Hib disease.

◇ Children at high risk for invasive *S. pneumoniae* disease include those with chronic heart disease (particularly cyanotic congenital heart disease and cardiac failure); chronic lung disease (including asthma if treated with high-dose oral corticosteroid therapy); diabetes mellitus; cerebrospinal fluid leak; cochlear implant; sickle cell disease and other hemoglobinopathies; anatomic or functional asplenia; HIV infection; chronic kidney failure; nephrotic syndrome; diseases associated with treatment with immunosuppressive drugs or radiation therapy, including malignant neoplasms, leukemias, lymphomas, and Hodgkin disease; solid organ transplantation; or congenital immunodeficiency. Please refer to the UpToDate topics on PCVs and pneumococcal polysaccharide vaccines for a discussion of full *S. pneumoniae* immunization in children at high risk for invasive *S. pneumoniae* disease.

Table 6: Type of drug reactions

Type of reaction	Common features
Non-allergic reaction	- Adverse effects (eg, diarrhea, vomiting, yeast vaginitis) - Family history of penicillin allergy but no personal history
Mild non-IgE-mediated reaction	- Maculopapular rash (with or without itching) - Medical record lists penicillin allergy but patient unaware of reaction
IgE-mediated reaction	- Anaphylaxis - Angioedema - Wheezing - Laryngeal edema - Hypotension - Hives/urticaria
Serious delayed reactions	- Toxic epidermal necrolysis (TEN) - Stevens-Johnson syndrome (SJS) - Drug reaction with eosinophilia and systemic symptoms/drug-induced hypersensitivity syndrome (DRESS/DiHS) - Other exfoliating dermatoses/erythroderma - Serum sickness-like reactions - Drug-induced cytopenias - Drug-induced kidney, hepatic, or other specific organ damage

IgE: immunoglobulin E.

Graphic 117212 Version 3.0

Table 7: Suggested approach to the choice of empiric parenteral therapy for the treatment of suspected *Staphylococcus aureus* bacteremia and other invasive infections in infants and children older than 28 days

Clinical scenario	Antimicrobial regimens	
	Our preferred regimen	Alternate regimen(s)
Life-threatening infection suspected to be caused by <i>S. aureus</i> (health care- or community-associated)		
- Health care-associated or community-associated: - Sepsis - CNS infection - Endocarditis - Endovascular infection - Infection complicated by venous thrombosis	- Vancomycin 15 mg/kg IV every 6 hours (maximum daily dose 4 g/day)*, plus - Nafcillin or oxacillin 37.5 to 50 mg/kg IV every 6 hours (maximum daily dose 12 g/day)	- One of the following: - Ceftaroline^Δ 15 mg/kg IV every 8 hours (maximum dose 600 mg) - Linezolid^Δ <12 years: 30 mg/kg per day IV in 3 doses ≥12 years: 600 mg twice per day IV - Daptomycin^{Δ◇} (for patients without pneumonia) 1 through 6 years: 12 mg/kg per day IV once daily 7 through 11 years: 9 mg/kg per day IV once daily 12 through 17 years: 7 mg/kg per day IV once daily
Suspected health care-associated or community MRSA pneumonia complicating influenza[¶]	- Vancomycin, plus - A second anti-MRSA agent (eg, clindamycin, ceftaroline^Δ, linezolid^Δ), plus - Either nafcillin or oxacillin, plus - Anti-influenza therapy	- One of the following: - Ceftaroline^Δ 15 mg/kg IV every 8 hours (maximum dose 600 mg) - Linezolid^Δ <12 years: 30 mg/kg per day IV in 3 doses ≥12 years: 600 mg twice per day IV Plus - Anti-influenza therapy

Nonlife-threatening, nonendovascular infection (eg, pneumonia, septic arthritis, osteomyelitis) without signs of sepsis suspected to be caused by *S. aureus*

Health care-associated[§]	Vancomycin 15 mg/kg IV every 6 to 8 hours (maximum daily dose 4 g/day)*	- One of the following: - Ceftaroline^Δ 15 mg/kg IV every 8 hours (maximum dose 600 mg) - Linezolid^Δ <12 years: 30 mg/kg per day IV in 3 doses ≥12 years: 600 mg twice per day IV - Daptomycin^{Δ◇} (for patients without pneumonia) 1 through 6 years: 12 mg/kg per day IV once daily 7 through 11 years: 9 mg/kg per day IV once daily 12 through 17 years: 7 mg/kg per day IV once daily
- Community-associated >10% of <i>S. aureus</i> isolates are MRSA[¥] - Prevalence of clindamycin resistance ≥15%[‡]	Vancomycin 15 mg/kg IV every 6 to 8 hours (maximum daily dose 4 g/day)*	- One of the following: - Ceftaroline^Δ 15 mg/kg IV every 8 hours (maximum dose 600 mg) - Linezolid^Δ <12 years: 30 mg/kg per day IV in 3 doses ≥12 years: 600 mg twice per day IV - Daptomycin^{Δ◇} (for patients without pneumonia) 1 through 6 years: 12 mg/kg per day IV once daily 7 through 11 years: 9 mg/kg per day IV once daily 12 through 17 years: 7 mg/kg per day IV once daily
Community-associated	Clindamycin 40 mg/kg per day IV divided in 3 or 4 doses	One of the following:

▪ >10% of <i>S. aureus</i> isolates are MRSA[¥] ▪ Prevalence of clindamycin resistance <15%[‡] ▪ Bacteremia unlikely	(maximum daily dose 2.7 g/day)	• Vancomycin 15 mg/kg IV every 6 to 8 hours (maximum daily dose 4 g/day)[*] • Ceftaroline^Δ 15 mg/kg IV every 8 hours (maximum dose 600 mg) • Daptomycin^{Δ◇} (for patients without pneumonia) ◦ 1 through 6 years: 12 mg/kg per day IV once daily ◦ 7 through 11 years: 9 mg/kg per day IV once daily ◦ 12 through 17 years: 7 mg/kg per day IV once daily
▪ Community-associated ▪ ≤10% of <i>S. aureus</i> isolates are MRSA[¥]	▪ Nafcillin or oxacillin 150 to 200 mg/kg per day IV in 4 to 6 doses (maximum daily dose 12 g/day)	▪ One of the following: • Cefazolin 100 to 150 mg/kg per day IV in 3 doses • Vancomycin 15 mg/kg IV every 6 to 8 hours (maximum daily dose 4 g/day)[*]

This table is meant to be used in conjunction with UpToDate content on *S. aureus* infections in children. In most cases, invasive *S. aureus* infection is suspected because there are clinical findings suggestive of a clinical syndrome or focal source that is associated with *S. aureus* infection (eg, osteomyelitis, septic arthritis, intravascular catheter). Less commonly, *S. aureus* is suspected because of underlying risk factors or Gram stain demonstrating gram-positive cocci in clusters. Additional antibiotics may be required for nonstaphylococcal pathogens until the pathogen is identified. Final therapy decisions are based on results of cultures and susceptibility testing. Refer to UpToDate content for additional information about choice of antimicrobial therapy. Consultation with an expert in infectious diseases may be warranted for guidance regarding choice and duration of antimicrobial therapy.

AUC: area under the curve; CNS: central nervous system; IV: intravenous; MRSA: methicillin-resistant *S. aureus*.

* Alternative dosing is suggested for clinicians/institutions who follow AUC-guided therapeutic monitoring for vancomycin for serious MRSA infections as suggested by [consensus guidelines](#)^[4]; this strategy requires input from a clinical pharmacist, who will provide recommendations for initial dosing. Refer to UpToDate content on invasive staphylococcal infections in children for details of trough-guided and AUC-guided vancomycin dosing.

¶ For suspected MRSA pneumonia complicating influenza, addition of a second anti-MRSA agent (eg, clindamycin, ceftaroline, linezolid) within the first 24 hours of admission may be associated with

decreased mortality. Refer to UpToDate content on treatment of invasive *S. aureus* infections in children for details.

Δ Experience with these agents in children is limited. Consultation with an expert in infectious diseases may be warranted.

◇ Daptomycin should not be used in children with concomitant pulmonary involvement. Daptomycin is active in vitro against multidrug-resistant gram-positive organisms, including *S. aureus*, but is not well studied in children. It is approved by the US Food and Drug Administration for the treatment of complicated skin and skin-structure infections in patients ≥1 year of age, the treatment of *S. aureus* bacteremia in children 1 through 17 years of age, and the treatment of *S. aureus* bacteremia (including right-sided endocarditis) in patients ≥18 years of age. Dosing for other indications is not well established. The dosing guidance for daptomycin provided in the table above is based upon the package label.^[5] Limited clinical data suggest that higher doses may be needed to effectively treat *S. aureus* bacteremia in children (at least 20 mg/kg in children ≤6 years, 17 mg/kg in children 7-11 years, and 10 mg/kg in adolescents ≥12 years).^[6]

§ Risk factors for health care-associated infection include personal history of or frequent contact with an individual with a history of hospitalization, surgery, dialysis, or residence in a long-term care facility within the previous year, frequent contact with the health care environment, presence of an invasive device (eg, intravascular catheter or tracheal tube), and history of MRSA infection or colonization.

¥ Information regarding local susceptibility patterns can be obtained from local public health officials or hospital laboratories. We provide coverage for MRSA when >10% of community isolates are MRSA; other experts may use a different threshold.

‡ Information regarding local susceptibility patterns can be obtained from local public health officials or hospital laboratories. The threshold prevalence of clindamycin-resistant MRSA for choosing vancomycin varies from center to center, usually ranging from 10 to 25%, in an effort to balance the benefit of definitive therapy for the patient with the risk of increasing vancomycin resistance in the community. Additional considerations in the decision to choose vancomycin include the prevalence of MRSA in the community, the severity of illness, and the turn-around time for susceptibilities.

References:

1. American Academy of Pediatrics. *Staphylococcus aureus*. In: *Red Book: 2024-2027 Report of the Committee on Infectious Diseases*, 33rd ed, Kimberlin DW, Banerjee R, Barnett ED, Lynfield R, Sawyer MH (Eds), American Academy of Pediatrics, Itasca, IL 2024. p.767.
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Table 8: Alternative approaches to vancomycin dosing for children and adolescents with normal kidney function

Strategy	Initial dose and interval	Subsequent dose and interval adjustments
Traditional approach for children >1 month and adolescents	Typically 15 mg/kg per dose IV every 6 to 8 hours ^[1] : - Use the every-6-hour interval for serious infections* - Maximum daily dose: 4 g/day	Either: - Continue initial dose (for most children, particularly if duration of vancomycin is expected to be <3 days)[¶] - Based on trough-guided serum concentration monitoring for select children (eg, those with kidney dysfunction, infective endocarditis, risk factors for altered vancomycin kinetics [eg, fluid overload, critical illness])
AUC-guided approach ^{Δ[2]}	Generally requires consultation with a clinical pharmacist	Based on AUC-guided serum concentration monitoring (generally requires consultation with a clinical pharmacist)

The approach to vancomycin dosing is generally determined at the institutional level. Refer to UpToDate content on invasive staphylococcal infections in children for details of trough-guided and AUC-guided vancomycin dosing and traditional dosing of vancomycin for neonates.

AUC: area under the curve; IV: intravenous; MRSA: methicillin-resistant *Staphylococcus aureus*.

* Serious infections may include, but are not limited to, infective endocarditis, pneumonia requiring hospitalization, osteomyelitis, central nervous system infection, and infection causing critical illness.

¶ The value of trough-monitoring before achieving steady state (usually on day 2 to 3 of treatment) is uncertain.

Δ Some experts suggest this approach for serious* MRSA infections in children of all ages.

References:

- Kimberlin DW, Banerjee R, Barnett ED, et al (Eds). Tables of antibacterial drug dosages. In: Red Book: 2024-2027 Report of the Committee on Infectious Diseases, 33rd ed, American Academy of Pediatrics 2024.
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Table 9: Important aspects of the history in a child with pneumonia

Historical feature	Possible significance
Age of the child	Viral etiologies are most common in infants and preschool children Atypical bacterial pathogens are more common in school-age children
Recent viral upper respiratory tract infection	May predispose to bacterial superinfection with <i>Streptococcus pneumoniae</i> or <i>Staphylococcus aureus</i>
Associated symptoms	<i>Mycoplasma pneumoniae</i> is often associated with extrapulmonary manifestations (eg, headache, sore throat, conjunctivitis, photophobia, rash)
Cough, chest pain, shortness of breath, difficulty breathing	"Classic" features of pneumonia, but nonspecific
Increased work of breathing in the absence of stridor or wheezing	Suggestive of more severe pneumonia
Fluid and nutrition intake	Difficulty or inability to feed suggests severe illness
Choking episode	May indicate foreign body aspiration
Duration of symptoms	Chronic cough (>4 weeks) suggests etiology other than acute pneumonia (refer to UpToDate topic on causes of chronic cough in children)
Previous episodes	Recurrent episodes may indicate aspiration, congenital or acquired anatomic abnormality, cystic fibrosis, immunodeficiency, asthma, missed foreign body
Immunization status	Completion of the primary series of immunizations for <i>Haemophilus influenzae</i> type b, <i>S. pneumoniae</i> , <i>Bordetella pertussis</i> , and seasonal influenza decreases, but does not eliminate, the risk of infection with these organisms
Previous antibiotic therapy	Increases the likelihood of antibiotic-resistant bacteria
Maternal history of chlamydia during pregnancy (for infants <4 months of age)	May indicate <i>Chlamydia trachomatis</i> infection
Foreign travel or known exposure to tuberculosis	May indicate <i>Mycobacterium tuberculosis</i> infection
Ill contacts	More common with viral etiologies
Travel to or residence in certain areas that suggest endemic pathogens	- Measles: Resource-limited countries - Coccidioidomycosis: Southwestern United States, northern Mexico, Central and South America - Blastomycosis: Southeastern and Central United States; states bordering the Great Lakes

	▪ Histoplasmosis: Ohio, Missouri, and Mississippi River valleys in the United States; Canada; Central America; eastern and southern Europe; parts of Africa; eastern Asia; and Australia ▪ Hantavirus: West of the Mississippi River; four corners region of United States (where borders of Colorado, New Mexico, Arizona, and Utah meet) ▪ Middle East respiratory syndrome (MERS): Countries in or near Arabian Peninsula
Animal exposure	May indicate histoplasmosis, psittacosis, Q fever
Day care center attendance	Exposure to viruses and antibiotic-resistant bacteria

Graphic 52510 Version 15.0

Contributor Disclosures

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